



Neonatal and Fetal Conditions

Potential survival benefit with repair of congenital diaphragmatic hernia (CDH) after extracorporeal membrane oxygenation (ECMO) in select patients: Study by ELSO CDH Interest Group

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ABSTRACT

Purpose: Studying the timing of repair in CDH is prone to confounding factors, including variability in disease severity and management. We hypothesized that delaying repair until post-ECMO would confer a survival benefit.

Methods: Neonates who underwent CDH repair were identified within the ELSO Registry. Patients were then divided into on-ECMO versus post-ECMO repair. Patients were 1:1 matched for severity based on pre-ECMO covariates using the propensity score (PS) for the timing of repair. Outcomes examined included mortality and severe neurologic injury (SNI).

Results: After matching, 2,224 infants were included. On-ECMO repair was associated with greater than 3-fold higher odds of mortality (OR 3.41, 95% CI: 2.84–4.09, $p < 0.01$). The odds of SNI was also higher for on-ECMO repair (OR 1.49, 95% CI: 1.13–1.96, $p < 0.01$). A sensitivity analysis was performed by including the length of ECMO as an additional matching variable. On-ECMO repair was still associated with higher odds of mortality (OR 2.38, 95% CI: 1.96–2.89, $p < 0.01$). Results for SNI were similar but were no longer statistically significant (OR 1.33, 95% CI: 0.99–1.79, $p = 0.06$).

Conclusions: Of the infants who can be liberated from ECMO and undergo CDH repair, there is a potential survival benefit for delaying CDH repair until after decannulation.

Type of Study: Treatment Study

Level of Evidence: III

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1. Introduction

Since the first description of CDH treatment with ECMO in 1977 by Drs. German and Bartlett, refinement of this technique has led to improved outcomes [1]. But, over the last 2 decades, the mortality for CDH-neonates who are treated with ECMO has remained relatively unchanged at 50% [2]. While some of this mortality is attributable to disease severity and patient factors, addressing the issue of when to

operate is pivotal to understanding surgeon- or ECMO team-modifiable factors that can alter the outcome for these infants. The decision to repair during an ECMO run (on-ECMO) vs. after the completion of an ECMO run (post-ECMO) is typically a discrete choice by the treating physician, institution, or ECMO team. Without robust data to guide this decision, any anecdotally driven choice is subject to potential individual and institutional bias. The one exception to this is repair before ECMO (pre-ECMO), in which situation the need for ECMO is typically an unexpected event most likely secondary to postoperative disease progression.

Proponents of on-ECMO repair believe that repair of the diaphragmatic defect may lead to additional lung recruitment, increased thoracic space, and minimization of the risk of an unexpected second run due to peri-operative stress. However, the anticoagulation required for ECMO

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and subsequent risk of bleeding is the primary disadvantage of this strategy and could even require premature decannulation. The timing of repair on-ECMO (early vs. late) is another complex issue with reportedly good results that vary between institution [3–7]. However, those that oppose this strategy argue that repair does not necessarily help with the management of pulmonary hypertension and that post-ECMO repair is easier without the risk of anticoagulation – if patients are doing well and will successfully wean from ECMO, why not delay the operation until it is potentially safer? Unfortunately, early prediction of severity and ECMO course, although recently studied, remains challenging [8].

Currently, there are no randomized controlled trials (RCT) to address timing of repair and development of such a trial is problematic because of the significant variability in disease severity, which can potentially lead to selection bias. Many of the studies addressing this topic are single institutional experiences; thus we sought to better understand the effect that timing of repair in relation to ECMO would have on mortality in a larger dataset. By using propensity score (PS) matching, we hoped to simulate the study conditions of an RCT on retrospective data by using the Extracorporeal Life Support Organization (ELSO) registry database, which has not been utilized for this topic since 1994. At that time, they concluded that repair on-ECMO was associated with increased bleeding complications [9]. We hypothesized that repair on-ECMO as opposed to post-ECMO would be associated with a significant increase in mortality and major complications, specifically severe neurologic injury (SNI).

2. Methods

2.1. Data source and cohort

This study was approved by the Children's Hospital of Orange County institutional review board (#150969) and the Extracorporeal Life Support Organization (ELSO) Scientific and Oversight Committee. The ELSO Registry, which collects clinical information for adults and children treated with ECMO, was queried for neonates whose primary diagnosis was CDH from 2000–2016 and who underwent CDH repair either during or after ECMO. In those neonates that received multiple ECMO runs, the first ECMO run was used.

2.2. Outcomes, treatment variables and analysis cohorts

The main a priori specified exposure was timing of repair in relation to ECMO run. We dichotomized infants into those who underwent repair on-ECMO and those that underwent repair post-ECMO. The prespecified primary and secondary outcomes were mortality at discharge and SNI, respectively. SNI was defined as acute neurologic events during ECMO represented by the ELSO registry – CNS hemorrhage, infarct and/or intraventricular hemorrhage (IVH) grade 3 and 4. Cerebral infarction or intracerebral hemorrhage (ICH) reported to the ELSO Registry are diagnosed using head ultrasonography or computerized tomography [10]. We excluded seizure and grades 1&2 IVH from the definition of SNI.

2.3. Statistical analysis

Propensity score (PS) matching [11] was performed to select the matched study cohorts. Controls (neonates who underwent on-ECMO repair) were matched to cases (neonates who underwent post-ECMO repair) based on the estimated PS given his/her baseline pre-ECMO covariates. A logistic regression model was used to estimate propensity scores using a response variable corresponding with timing of repair (1 = post-ECMO repair, 0 = on-ECMO repair), in conjunction with pre-ECMO covariates: demographics variables including gender, pre-ECMO weight, race, gestational age (GA), post-gestational age, 5min Apgar, side of CDH, prenatal diagnosis of CDH, hand-bagging and

pre-ECMO arrest; blood gas/ventilator variables included pH, pCO₂ and pO₂, mean airway pressure (MAP), oxygenation index (OI); pre-ECMO rescue therapies included inotropes, bicarbonate/tromethamine (THAM), inhaled nitric oxide (iNO), surfactant, neuromuscular blockers, milrinone, sildenafil and steroids; comorbidity variables included critical congenital heart disease (CCHD), multiple congenital anomalies (MCA), chromosomal anomalies, perinatal infection. Missing values of birth weight, gestational age, pCO₂, pO₂ and OI were obtained using mean imputation. PS matching criteria was targeted at 1:1 (with radius = 0.15*SD of PS). The success of PS matching to obtain comparable analysis cohorts was checked by comparing the pre-ECMO covariates before and after matching based on their standardized differences. Also, descriptive summary of continuous pre-ECMO variables were reported as mean with standard deviation (SD) and compared using two-sample t-test. Categorical variables were compared using chi-square test before and after PS matching.

A priori specified main analyses of the outcomes were univariate logistic regression models based on the matched sets for each cohort. A sensitivity analysis was also performed in which length of ECMO run was included in the PS matching. Results were reported as odds ratio (OR) with 95% confidence interval (CI). Analyses were performed using SAS version 9.3 and R version 3.2.2.

3. Results

3.1. Unmatched baseline characteristics

We identified 3,045 infants between 2000–2016 who were treated with ECMO and who underwent CDH repair. Of these 1,817 (59.7%) underwent repair on-ECMO while 1,228 (40.3%) underwent repair post-ECMO. The observed mortality rates for the on-ECMO and post-ECMO repair groups were 54.4% and 22.7%, respectively. The incidence of SNI was 12.6% for the on-ECMO repair group and 8.7% for the post-ECMO repair group. During this period, there were also 741 infants who did not undergo repair and were excluded from the primary analysis. The observed mortality of these infants was 92.7% and the incidence of SNI was 32.3%.

There were multiple significant differences between the baseline characteristics of each of the two groups prior to matching including demographics, blood gas/ventilator, and pre-ECMO rescue therapies (Table 1). However, there were no statistically significant differences after PS matching (Table 1, Figure 1).

The on-ECMO repair group also had longer ECMO runs with the average being 2.1 weeks as opposed to 1.4 weeks in the post-ECMO repair.

Among those infants that had more than one ECMO run (N = 99), 75.8% underwent on-ECMO repair and 12.7% underwent post-ECMO repair. The rate of second run of ECMO was 2 times more likely for the on-ECMO repair group (1.95% vs. 4.13%, $p < 0.01$).

3.2. Propensity score matched cohorts

3.2.1. Primary outcome: mortality

The PS matching identified 2,225 neonates (1,112 for each treatment group). Table 1 and Figure 1A show that there were no significant differences in baseline characteristics, blood gas/ventilatory settings, pre-ECMO therapies, or comorbidity variables after PS matching. The odds of mortality was significantly higher in the on-ECMO repair group when compared to the post-ECMO repair group (OR: 3.41, 95% CI: 2.843–4.094, $P < 0.0001$, Table 2a). Sensitivity analysis using logistic regression with the inclusion of estimated propensity score in a matched set as an additional covariate yielded the same conclusion with respect to all outcomes.

As an additional sensitivity analysis, we performed a secondary analysis in which we included ECMO length as part of the PS matching to further account for a potentially concealed effect of subject severity.

Table 1
Demographics, blood gas/ventilator, pre-ECMO rescue therapies and comorbidity variables for those neonates who underwent on-ECMO and post-ECMO repair before and after propensity score matching.

Parameter	Category	Before matching (N = 3,045)				After matching (N = 2,224)			
		On-ECMO Repair (N = 1,817) N (%) / Mean (SD)	Post-ECMO Repair (N = 1,228) N (%) / Mean (SD)	Standardized difference	p Value	On-ECMO Repair (N = 1112) N (%) / Mean (SD)	Post-ECMO Repair (N = 1112) N (%) / Mean (SD)	Standardized difference	p Value
Sex	Female	769 (42.32)	511 (41.61)	0.0144	0.6970	471 (42.36)	475 (42.72)	-0.0073	0.8638
	Male	1048 (57.68)	717 (58.39)			641 (57.64)	637 (57.28)		
Birthweight		3.05 (0.49)	3.11 (0.48)	0.1395	0.0002	3.09 (0.48)	3.09 (0.48)	0.0010	0.9814
Race	Whites	1154 (63.51)	695 (56.60)	0.1760	<0.0001	658 (59.17)	650 (58.45)	0.0175	0.9823
	Hispanics	275 (15.13)	265 (21.58)			211 (18.97)	217 (19.51)		
	Blacks	223 (12.27)	156 (12.70)			138 (12.41)	141 (12.68)		
	Others	165 (9.08)	112 (9.12)			105 (9.44)	104 (9.35)		
Gestational age		38.13 (1.64)	38.27 (1.65)	0.0896	0.0153	38.23 (1.60)	38.22 (1.66)	-0.0092	0.8284
Apgar at 5 mins		6.07 (1.97)	6.46 (1.91)	0.1965	<0.0001	6.38 (1.90)	6.39 (1.93)	0.0032	0.9397
Post gestational Age (days)		1.60 (2.63)	1.78 (2.73)	0.0648	0.0785	1.70 (2.59)	1.77 (2.77)	0.0245	0.5636
Side of diaphragmatic hernia	Left	1385 (76.22)	903 (73.53)	0.1101	0.0330	828 (74.46)	831 (74.73)	0.0279	0.9331
	Right	387 (21.30)	307 (25.00)			262 (23.56)	263 (23.65)		
	Both	18 (0.99)	7 (0.57)			8 (0.72)	7 (0.63)		
	Missing	27 (1.49)	11 (0.90)			14 (1.26)	11 (0.99)		
Prenatal diagnosis		1283 (70.61)	775 (63.11)	-0.1598	<0.0001	713 (64.12)	726 (65.29)	0.0245	0.5641
Hand bagging	No	1662 (91.47)	1189 (96.82)	0.2359	<0.0001	1069 (96.13)	1073 (96.49)	0.0201	0.8939
	Yes	122 (6.71)	26 (2.12)			28 (2.52)	26 (2.34)		
	Missing	33 (1.82)	13 (1.06)			15 (1.35)	13 (1.17)		
Patient arrested before ECMO		137 (7.54)	62 (5.05)	-0.1027	0.0064	62 (5.58)	61 (5.49)	-0.0039	0.9261
pH		7.15 (0.17)	7.19 (0.15)	0.2559	<0.0001	7.19 (0.16)	7.19 (0.15)	-0.0222	0.6003
pco2		71.45 (28.25)	65.58 (25.73)	-0.2172	<0.0001	65.93 (24.51)	66.63 (26.22)	0.0274	0.5187
po2		38.00 (21.85)	38.27 (23.71)	0.0119	0.7454	38.45 (23.94)	38.20 (24.02)	-0.0103	0.8085
HFOV		1334 (73.42)	926 (75.41)	0.0456	0.2183	858 (77.16)	835 (75.09)	-0.0485	0.2526
MAP		16.43 (3.99)	16.26 (4.18)	-0.0419	0.2549	16.45 (3.68)	16.32 (4.26)	-0.0323	0.4463
Oxygenation index		55.02 (33.91)	53.08 (35.21)	-0.0561	0.1273	53.97 (33.13)	53.62 (35.78)	-0.0100	0.8139
Inotropes		1584 (87.18)	1123 (91.45)	0.1386	0.0002	1013 (91.10)	1008 (90.65)	-0.0156	0.7128
Bicarbonate/THAM		577 (31.76)	452 (36.81)	0.1066	0.0038	392 (35.25)	386 (34.71)	-0.0113	0.7896
Nitric oxide		1459 (80.30)	1068 (86.97)	0.1811	<0.0001	971 (87.32)	955 (85.88)	-0.0422	0.3193
Surfactant		282 (15.52)	224 (18.24)	0.0727	0.0478	211 (18.97)	198 (17.81)	-0.0302	0.4767
Neuromuscular blockers		1063 (58.50)	764 (62.21)	0.0759	0.0403	669 (60.16)	685 (61.60)	0.0295	0.4869
Milrinone		130 (7.15)	129 (10.50)	0.1183	0.0012	102 (9.17)	103 (9.26)	0.0031	0.9416
Sildenafil		15 (0.83)	16 (1.30)	0.0465	0.1980	13 (1.17)	13 (1.17)	0.0000	1.0000
Steroids		104 (5.72)	125 (10.18)	0.1652	<0.0001	92 (8.27)	96 (8.63)	0.0129	0.7604
CCHD		33 (1.82)	16 (1.30)	-0.0414	0.2695	16 (1.44)	14 (1.26)	-0.0156	0.7131
MCA		9 (0.50)	2 (0.16)	-0.0581	0.1336	1 (0.09)	2 (0.18)	0.0245	0.5634
Chromosomal		19 (1.05)	10 (0.81)	-0.0241	0.5191	7 (0.63)	10 (0.90)	0.0310	0.4651
Perinatal infection		32 (1.76)	26 (2.12)	0.0258	0.4807	23 (2.07)	22 (1.98)	-0.0064	0.8803

Figure 1B demonstrates that after matching there were no significant differences in baseline characteristics, blood gas/ventilatory settings, pre-ECMO therapies or comorbidity variables after PS matching. The odds of mortality remained elevated in the on-ECMO repair group when compared to the post-ECMO repair group (OR: 2.38, 95% CI: 1.959–2.888, $P < 0.0001$, Table 2b).

3.2.2. Secondary outcome: SNI

There was significantly increased odds of SNI in the on-ECMO repair group when compared to the post-ECMO repair group (OR: 1.49, 95% CI: 1.129–1.958, $P = 0.005$, Table 2a). However, when length of ECMO run was included in the PS-matching, this association did not maintain statistical significance (OR: 1.33, 95% CI: 0.987–1.788, $P = 0.0607$).

4. Discussion

The results of our study indicate that infants that can be successfully liberated from ECMO and undergo post-ECMO repair have an improved survival compared with those repaired while on ECMO. This supports our hypothesis that repair on-ECMO is associated with an increased mortality. Our data seems consistent with several other publications. In 2009, the CDH Study Group published a retrospective review of

636 patients showing a hazard ratio of 1.4 for on-ECMO repair [12]. Though the reported effect is more modest, and despite recognized limitations, it agrees with the findings of our study. It is possible that a larger dataset and using PS matching instead of Cox regression allowed us to better evaluate the true effect of this single treatment variable on mortality. Robertson et al., in a single institution study published in 2017, suggested that delaying repair until after decannulation was associated with both improved survival and a shorter period on ECMO [7]. Also in 2017, the Children's Hospital of Los Angeles published similar findings; infants that were decannulated prior to repair had shorter ECMO runs and improved survival [13]. Given the totality of the evidence reported by multiple groups, the Canadian CDH Collaborative recommended delaying diaphragm repair until decannulation [14]. To our knowledge, our study represents the largest cohort of CDH-ECMO patients addressing the issue of timing of repair, and it demonstrates a clear advantage to delaying repair until the completion of the ECMO run, if possible.

We also showed that on-ECMO repair was associated with longer ECMO runs and that those infants were more likely to require a second run. This is similar to the conclusion that Partridge et al. reached when comparing on vs. post-ECMO repair at the Children's Hospital of Philadelphia [6]. And it may be this effect that is contributing to the

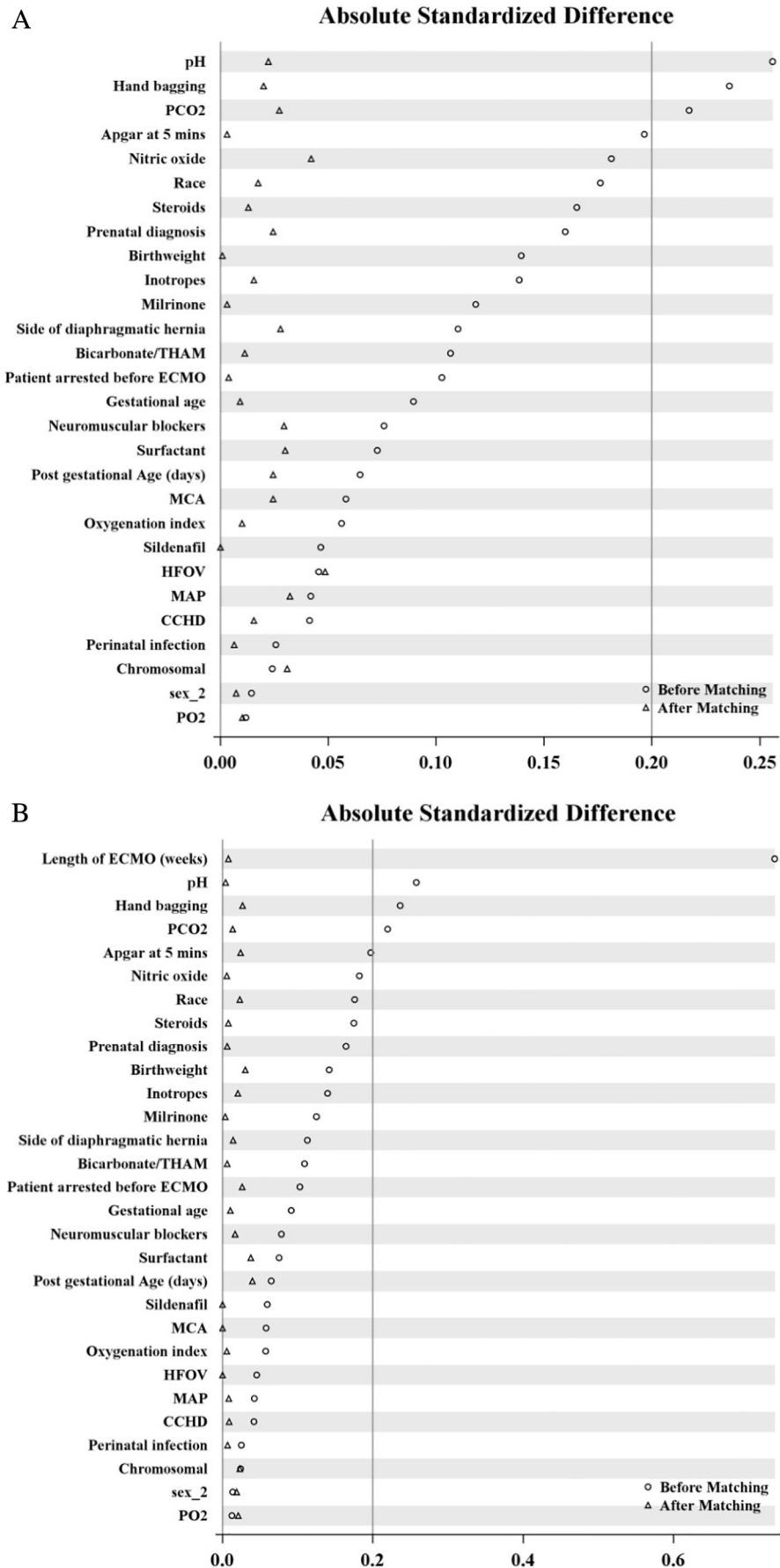


Figure 1. Plot of standardized difference of all covariates before and after matching: A) all infants B) sensitivity analysis including ECMO length.

Table 2

Odds ratio of mortality and severe neurologic injury for on-ECMO and post-ECMO repair: a) excluding ECMO run length
b) sensitivity analysis including ECMO run length

a)

Mortality				Severe Neurologic Injury			
Category	OR	95% CI	P Value	Category	OR	95% CI	P Value
Post-ECMO Repair		1.00 (Reference)		Post-ECMO Repair		1.00 (Reference)	
On-ECMO Repair	3.412	(2.843–4.094)	<0.0001	On-ECMO Repair	1.487	(1.129–1.958)	0.0047

b)

Mortality				Severe Neurologic Injury			
Category	OR	95% CI	P Value	Category	OR	95% CI	P Value
Post-ECMO Repair		1.00 (Reference)		Post-ECMO Repair		1.00 (Reference)	
On-ECMO Repair	2.378	(1.959–2.888)	<0.0001	On-ECMO Repair	1.329	(0.987–1.788)	0.0607

increased rates of SNI that were seen with the on-ECMO repair group. We did not initially include ECMO length in PS matching because, like an RCT, this information would not be known at the start of study. We elected to perform a sensitivity analysis including ECMO run length in PS matching as a potential surrogate for disease severity. Though the effect on mortality was similar, it mitigated the increased odds of SNI.

Our conclusions are limited to the comparison of two groups: all infants repaired on-ECMO vs. all infants repaired after ECMO. We cannot draw conclusions regarding the timing of repair on-ECMO, nor subsets of that group (e.g. early on-ECMO repair) to post-ECMO repair. The APSA guidelines committee suggested that early repair on ECMO is potentially associated with a reduction in mortality and shorter ECMO runs [15]. This is largely based on two single institution studies published by Dassinger et al. and Fallon et al [3,16]. Groups from Texas and Arkansas showed an overall survival of around 70% in infants repaired early on ECMO after a specific change to their institutional protocol. Those results are better than the historic 50% survival rate, and the 54% mortality that we observed in the on-ECMO repair group [2]. In contrast, our institutional data collected from the ELSO registry showed a 77% raw survival rate for infants that were repaired post-ECMO. Unfortunately, we were unable to account for timing of repair on ECMO. While ELSO implemented a system for collecting this data more rigorously in 2016, its reporting is inconsistent prior to that time. This is potentially problematic as the infant that is stable enough on ECMO to be repaired within 72 hours is a very different patient than an infant that is being offered a salvage repair due to inability to wean from ECMO. We attempted to address this issue by including ECMO length in the PS matching, and the survival benefit to post-ECMO repair remained.

The PS matching design has certain advantages as it is better able to address the effect of a single treatment variable on an outcome with a retrospective cohort of patients. However, it is limited to the available variables and cannot account for unknown confounders, which can only be addressed with a RCT design. Perhaps the largest limitations are the inability to address center/provider specific variables and the inability to address the timing of repair on-ECMO. Individual centers have shown good results with early repair on ECMO, while others have not. We cannot address why this variability would exist with the data available to us. There were also several other variables that are missing from the ELSO dataset, such as prenatal imaging (e.g. observed/expected lung-head ratio, liver position) and defect size. These variables, particularly prenatal imaging, are not confounded by provider intervention and have been shown to be important for prognosis [17–20]. This could potentially limit the effectiveness of our PS matching.

Our work has several very important limitations. First, there was also a subset of patients (20%) in our dataset that never underwent repair and only 7% of them survived. In a typical RCT, an intent-to-treat analysis would include these infants in one of the two treatment groups (on-ECMO vs. post-ECMO). It is impossible with the data available to us

to know the practice patterns or intent of the treating physician for these infants, so they were excluded from our analysis. Further, we do not know how many patients with CDH were born but never offered ECMO (let alone repair). Those patients that are non-repaired are an important cohort to consider in CDH analyses. Due to limitations of our dataset, we could not include them. Second, despite a sophisticated analysis, all factors cannot be accounted for. Therefore, despite narrowing the difference in covariates and working to eliminate statistical differences based on the data collected, it is impossible to completely account for clinical differences which may exist. Third, prediction of risk among CDH patients in the early phase on ECMO remains an inexact science. Improved stratification may help to identify a low-risk sub-set of CDH patients on ECMO who may benefit from an attempt to delay repair versus a high-risk sub-set of CDH patients on ECMO who may benefit from an early repair. Our findings may inform tailored CDH management based on risk stratification.

This study is also subject to the typical limitations of retrospective database studies, including miscoding errors or treatment selection bias. Regarding specific variables, the ELSO registry reports the single worst value for pre-ECMO blood gas and ventilatory variables that are closest to ECMO initiation, as such we cannot account for trends in these variables prior to ECMO initiation. Another limitation of the ELSO registry is the selection bias of those infants that are included as they are all treated with ECMO, but, in this case, our comparison was limited to this subset of CDH-infants anyways.

In summary, there appears to be a potential survival benefit associated with delaying repair until after ECMO as opposed to repair on-ECMO. This is similar to the findings of other single institution and retrospective database studies. However, it would not be accurate to condemn all repair on-ECMO, and the correct timing becomes less clear in certain infants, especially those infants that cannot be successfully weaned who may have the severest form of the disease. The results of this study found that for those infants that can be successfully weaned from ECMO, repair should be delayed until after decannulation. Further studies, particularly from controlled trials, are needed to define the role of early repair of CDH on-ECMO compared to repair late on-ECMO and repair post-ECMO.

Discussion

Potential Survival Benefit with Repair of Congenital Diaphragmatic Hernia (CDH) After Extracorporeal Membrane Oxygenation (ECMO) in Select Patients: Study by ELSO CDH Interest Group

Dr. Russell: We had early data that said we should repair off-ECMO, and then we had a wave of data that said we should repair early on-ECMO. Now, we're seeing data again that says maybe we should repair off-ECMO with the caveat that you may have included a lot of patients that had the late repair on-ECMO, which we know is a disaster. But it does seem like the data is moving toward equipoise for the trial that

you're speaking of. How many people here fix early on-ECMO? And how many people wait until they're off-ECMO? There's your equipoise. Thank you.

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